

Math and Diagrams as Images

PDF has no client-side renderer, so math and diagram fences are turned into embedded raster images through the `renderers:` callables you supply. Without a renderer they degrade gracefully to their monospace source.

Inline and display math

The mass-energy relation $E = mc^2$ appears inline in a sentence, while the Gaussian integral is shown as a centered display equation:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Another inline example: the quadratic roots $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Diagrams

A flowchart, rendered through the `:mermaid` renderer callable:

```
[mermaid] graph TD; Parse --> AST --> Render --> PDF
```

A dependency graph, rendered through the `:graphviz` callable:

```
[graphviz] digraph {
  carve_rb -> carve_hexapdf;
```

Each callable receives the fence source and returns image bytes (PNG/JPG); the engine embeds the result. See `generate.rb` for a runnable renderer.